DAY 21 | POST-READ

Deploying to Azure: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
App Service	PaaS for web apps; Linux plan; deploy code or a container; set WEBSITES_PORT.
Container Apps	Serverless containers; az containerapp up --source; scales automatically, even to zero.
AKS and VMs	Full control, full responsibility. Choose only with a clear need.
CD pipeline	Test, build and push, deploy dev, approval, deploy prod with the same image tag.
Service connections	Scoped access from pipeline to Azure; prefer workload identity federation.
Variables vs secrets	Secrets are masked and must be mapped into env; best linked from Key Vault.
Key Vault (NEW v4)	Encrypted store for secrets, keys, certificates; RBAC roles; audit; rotation.
Managed identity (NEW v4)	Azure-issued identity for the app; short-lived tokens; no stored password.
Safe habits	Tag by build id, smoke test after deploy, roll back by tag, clean up.

Part B: Going beyond today

1. Key Vault references in App Service

The app still reads an environment variable; the value comes from Key Vault at runtime through the app's managed identity.

```
az webapp config appsettings set -g rg-demo -n tasks-api-demo123 --settings \
  "DATABASE_URL=@Microsoft.KeyVault(SecretUri=https://kv-demo.vault.azure.net/secrets/db-url/
```

2. Pulling from a private registry securely

Grant the app's managed identity the AcrPull role on the registry. No registry passwords are stored anywhere.

3. Key Vault in the pipeline

The AzureKeyVault task or a variable group linked to Key Vault fetches secrets at run time. Only map the secrets each step needs.

4. Deployment slots

App Service slots let you deploy to a staging slot, warm it up, and swap it with production almost instantly. Swapping back is your rollback.

5. Scale rules in Container Apps

```
az containerapp update -n tasks-api -g rg-demo --min-replicas 0 --max-replicas 5 \
  --scale-rule-name http --scale-rule-type http --scale-rule-http-concurrency 50
```

Part C: Check yourself

- Why should the same image be deployed to dev and prod?
- When would you choose Container Apps over App Service? When a VM?
- Where are deployment approvals configured in Azure DevOps?
- How does a managed identity remove the need for a stored password?
- How would you roll back a bad release?

Part D: Practice ideas for tonight

- Write a one-page runbook for SmartDesk: deploy, verify, roll back, rotate a secret, clean up.
- Add a smoke test stage to your pipeline YAML.
- List every secret SmartDesk uses and where each will live in Key Vault.

Coming up on Day 22: embeddings, vector databases, hybrid search and the RAG application.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today